



ALLIANCE
UNIVERSITY
Private University established in Karnataka State by Act No. 24 of year 2010
Recognized by the University Grants Commission (UGC), New Delhi.

NAAC
GRADE A+
ACCREDITED UNIVERSITY

Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 2

**Title: Sales Performance Analysis Across Products, Categories,
and Regions using Power BI Visualizations**

Submitted By:

Name: Kavivarshini S

Reg. No.: 2411021240004

Date: 31/08/2026

Submitted To:

Faculty Name: Mr. Aman Kumar Sharma

Department of Computer Application
Alliance University
Chandapura - Anekal Main Road, Anekal
Bengaluru – 562106

Sales Performance Analysis Across Products, Categories, and Regions using Power BI Visualization

Problem Statement

The main objective of this experiment is to analyze the sales performance of a retail business using Power BI. I used the Sample Superstore dataset to study sales and profit based on products, categories, regions, states, and customer segments. Different Power BI visualizations were created to make the business performance easier to understand.

Step 1: Dataset Collection

For this experiment, I used the **Sample Superstore dataset** from Kaggle. The dataset contains information about orders, customers, products, sales, profit, quantity, categories, regions, states, and dates. The dataset contains **9,994 records**.

The main fields used in my analysis are:

- Order ID
- Order Date
- Customer ID
- Product Name
- Category
- Sub-Category
- Region
- State
- Segment
- Sales
- Profit
- Quantity

Queries [4] < X ✓ fx = Table.TransformColumnTypes(#"Promoted headers", {{"Row ID", Int64.Type}, {"Postal Code", Int64.Type}},

Row ID	Order ID	Order Date	Ship Date	Ship Mode
1000 distinct, 1000 unique	479 distinct, 226 unique	373 distinct, 133 unique	398 distinct, 152 unique	4 distinct, 0 unique
1	CA-2016-152156	08-11-2016	11-11-2016	Second Class
2	CA-2016-152156	08-11-2016	11-11-2016	Second Class
3	CA-2016-138688	12-06-2016	16-06-2016	Second Class
4	US-2015-108966	11-10-2015	18-10-2015	Standard Class
5	US-2015-108966	11-10-2015	18-10-2015	Standard Class
6	CA-2014-115812	09-06-2014	14-06-2014	Standard Class
7	CA-2014-115812	09-06-2014	14-06-2014	Standard Class
8	CA-2014-115812	09-06-2014	14-06-2014	Standard Class
9	CA-2014-115812	09-06-2014	14-06-2014	Standard Class
10	CA-2014-115812	09-06-2014	14-06-2014	Standard Class
11	CA-2014-115812	09-06-2014	14-06-2014	Standard Class
12	CA-2014-115812	09-06-2014	14-06-2014	Standard Class
13	CA-2017-114412	15-04-2017	20-04-2017	Standard Class
14	CA-2016-161389	05-12-2016	10-12-2016	Standard Class
15	US-2015-118983	22-11-2015	26-11-2015	Standard Class
16	US-2015-118983	22-11-2015	26-11-2015	Standard Class
17	CA-2014-105893	11-11-2014	18-11-2014	Standard Class
18	CA-2014-167164	13-05-2014	15-05-2014	Second Class
19	CA-2014-143336	27-08-2014	01-09-2014	Second Class
20	CA-2014-143336	27-08-2014	01-09-2014	Second Class
21	CA-2014-143336	27-08-2014	01-09-2014	Second Class
22				

Step 2: Importing the Dataset

I imported the Sample Superstore dataset into **Power BI Desktop**. After importing the data, I checked the available fields and their values. The **Orders** table was used for creating the different visualizations because it contains the required sales, product, customer, region, and date information.

Step 3: Data Preparation

Before creating the dashboard, I checked the dataset using **Power Query Editor**.

The following checks were performed:

- Checked the column names.
- Verified the data types.
- Checked the Sales, Profit, and Quantity fields.
- Checked the Order Date and Ship Date fields.
- Checked for missing or incorrect values.
- Ensured that the required fields were ready for visualization.

This preparation helped to make the data suitable for further analysis.

Step 4: Creating the Visualizations

After preparing the data, I created different visualizations in Power BI to analyze the sales performance.

1. Total Sales

A card was created to display the total sales generated by the business. It provides a quick view of the overall sales performance.

2. Total Profit

A card was used to display the total profit. This helps in understanding the overall profitability of the business.

3. Total Quantity

A card was created to show the total quantity of products sold.

4. Customer Information

A customer-related card was included to display customer information from the dataset.

5. Sales by Region

A bar chart was created to compare sales across the different regions. This helps identify the regions with higher and lower sales.

6. Sales by Sub-Category

A bar chart was used to compare sales across different sub-categories. This helps identify the product groups that contribute more to sales.

7. Sales by Product

A product-wise sales chart was created to compare the sales generated by individual products.

8. Profit by Category

A column chart was created to compare profit among **Technology, Office Supplies, and Furniture**.

9. Sales by Segment

A pie chart was created to show the sales contribution of the three customer segments:

- Consumer

- Corporate
- Home Office

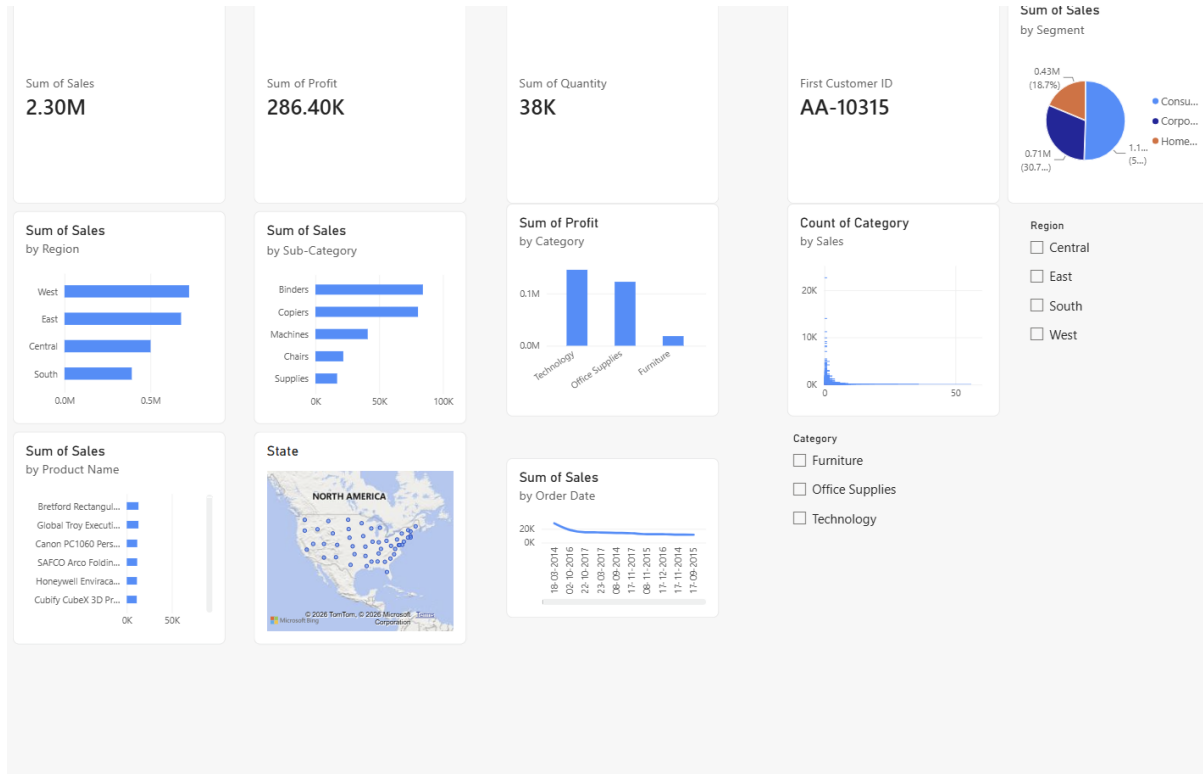
This helps to understand how sales are distributed among different customer groups.

10. Sales Trend

A line chart was created using **Order Date** and **Sales** to observe how sales changed over the period covered by the dataset.

11. Sales by State

A map was created using **State** and **Sales**. The bubbles on the map help identify states with higher sales.



Step 5: Adding Filters

I added **Region** and **Category** slicers to make the dashboard interactive. These slicers allow the user to select a particular region or category and view the corresponding sales information in the dashboard.

Step 6: Dashboard Observations

After analyzing the visualizations, I observed the following patterns:

- The **West region** shows strong sales performance among the four regions.
- Technology and Office Supplies show higher profit compared with Furniture.
- Sales vary between different sub-categories.
- Some products contribute more to overall sales than others.
- The Consumer, Corporate, and Home Office segments have different contributions to sales.
- Sales change at different points in the period shown in the line chart.

- The map shows that sales are distributed across different states, with some states having higher sales than others.

Business Use

The dashboard helps management get a quick understanding of sales performance across regions, products, categories, and customer segments. The information can be useful for identifying strong-performing areas and making decisions related to products, sales strategies, and regional performance.

Conclusion

In this experiment, I used Power BI to analyze the Sample Superstore dataset and create an interactive sales dashboard. Different visualizations such as cards, bar charts, a column chart, pie chart, line chart, and map were used to understand the business data.

The dashboard provides a simple way to view sales, profit, product performance, category performance, regional sales, and customer segment contributions. This experiment helped me understand how Power BI can be used to convert business data into useful visual information.

Learning Outcomes

After completing this experiment, I learned how to:

- Import a business dataset into Power BI.
- Prepare and check data using Power Query.
- Create KPI cards and different types of charts.
- Analyze sales based on regions, products, and categories.
- Compare profit across different categories.
- Use a pie chart to analyze customer segments.
- Create a map to visualize sales by state.
- Create a line chart to study sales trends.
- Use slicers to filter the dashboard.
- Interpret Power BI visualizations to identify business patterns.